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Python Strings

A string is a sequence of characters enclosed inside quotes (either double or single quotes). It is used to store and work with textual data in any language or symbol set supported by Unicode.

For example, `'Opus'` (same as `"Opus"`) is a string that contains characters `'O'`, `'p'`, `'u'` and `'s'`. By the way, these single characters are also strings themselves.

Python Multiline Strings

If we need to create a string containing multiple lines, we use triple double quotes `"""` or triple single quotes `'''`. For example,

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Output

```
To avoid pain, they avoid pleasure.
To avoid death, they avoid life.
```

Access String Characters

A string is a sequence, like a list or a tuple. This means characters in a string are ordered, and each one is associated with an integer, known as **index.**

Index of first character is **0**, index of second character is **1**, and so on.

```
model = 'ChatGPT'

# Access the first character
print(model[0])    # Output: C

# Access the fifth character
print(model[4])    # Output: G
```

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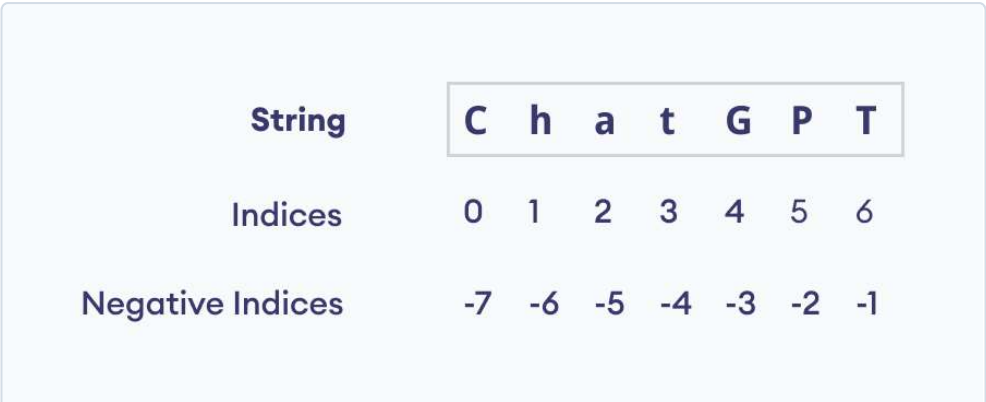
Similar to other sequences in Python, strings also support negative indexing. Index of last character is **-1**, index of second last character is **-2**, and so on.

```
model = 'ChatGPT'

# Access the last character
print(model[-1])   # Output: T

# Access the fourth last character
print(model[-4])   # Output: t
```

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String Indexing

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```
print(model[8])       # IndexError
```

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We can access a portion of a string (or any sequence) using colon `:` operator, known as slicing.

```
model = 'ChatGPT'

# Access characters from index 0 up to (but not including
print(model[0:4])   # Output: Chat
```

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You can learn more about it in Python slicing tutorial (/python-programming/slicing).

Strings are Immutable

In Python, strings are immutable; we cannot modify them. For example,

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Output

```
TypeError: 'str' object does not support item assignment
```

However, what we can do is create a modified version of the original string and assign it to a variable, without modifying the original string itself.

```
model = 'Opus'
version = '5'

model = model + " " + version
print(model)    # Opus 5
```

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In this program, we are concatenating strings `'Opus'`, `' '` and `'5'` using `+` operator to create a new string. Note that we are not modifying strings `'Opus'`, `' '` and `'5'`; that's not possible.

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```
text = "ChatGPT is great."

# Replace "ChatGPT" with "Claude"
new_text = text.replace("ChatGPT", "Claude")

print(new_text)  # Output: Claude is great.
```

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Here, `replace()` method returns a new string with `"ChatGPT"` replaced by `"Claude"`.

You can find a list of all string methods and their respective tutorials here (/python-programming/methods/string).

String Membership Test

We can test if a substring exists within a string or not using `in` keyword.

```
print('Chat' in 'ChatGPT')        # True
print('Claude' not in 'ChatGPT')  # True
```

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through it using a for loop (/python-programming/for-loop). For example,

```
model = 'Opus'

for c in model:
    print(c)
```

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Output

```
O
p
u
s
```

Python String Length

Besides string methods, Python has many built-in functions (/python-programming/methods/built-in) that provide different functionalities. One such commonly used function is len() (/python-programming/methods/built-in/len), which returns length (number of characters) of a string. For example,

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Escape Sequences

Escape sequences are used to include certain characters, such as quotes and newlines, inside a string.

Suppose a string has both single and double quotation marks as its characters. In such case, we can't do something like this:

```
example = "He said, "What's there?""

print(example) # Error
```

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Here, Python interpreter treats `"He said, "` as a string and doesn't know how to interpret the rest of the text. Hence, above code results in an error.

To solve this issue, we use escape character `\` in Python.

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```
print(example)

# Output: He said, "What's there?"
```

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Here are some commonly used escape sequences:

Escape Sequence	Description
`\\`	Backslash
`\'`	Single quote
`\"`	Double quote
`\b`	ASCII Backspace
`\n`	ASCII Linefeed
`\r`	ASCII Carriage Return
`\t`	ASCII Horizontal Tab

String Formatting (f-Strings)

Python f-strings (/python-programming/fstring) are used to create a new string by combining values and variables.

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```
company = 'Google'
field = 'AI'

message = f'{company} is an {field} company.'
print(message)
```

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Output

```
Google is an AI company.
```

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